



Climate Risk Intelligence

CRI Engine v0.3
Methodology Overview
Partnership Edition

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This document summarises the scientific foundations, data sources, and regulatory alignment of the CRI Engine. Proprietary calibration parameters, exact model weights, and implementation details are withheld pending a formal partnership agreement.

1. Executive Summary

The Climate Risk Intelligence (CRI) Engine is a quantitative framework that translates physical climate science and transition economics into asset-level financial risk metrics. It is designed to support regulatory disclosures under TCFD, IFRS S2, CSRD, the ECB's Guide on climate-related financial risks, and ASRS E1, while remaining actionable for investment and portfolio management decisions.

The engine assesses ten acute and chronic physical hazards, four transition risk channels, and three financial impact pathways across any asset with a known GPS coordinate or regional classification. Outputs are fully auditable — every number traces to a published peer-reviewed source or regulatory reference.

Key differentiators versus peer platforms:

- End-to-end regulatory alignment with IPCC AR6, TCFD, IFRS S2, CSRD, and EBA Pillar 3 simultaneously — not retrofitted.
- GPS-level precision via spatial downscaling from 25km climate grids to exact asset coordinates — no grid-cell approximation.
- Live meteorological baseline fetched from ERA5 reanalysis at each asset's coordinates, updated on-demand.
- Transparent audit trail: every hazard score, every data source, every formula linkage is disclosed in the output.
- Designed for financial professionals: outputs map directly to CaR (Climate at Risk), EV impact, and cost-of-capital adjustment.

2. Assessment Framework

2.1 Three-Pillar Structure

Every company or portfolio assessed through the CRI Engine is evaluated across three interconnected pillars. The composite Climate Rating reflects the weighted average of all three.

Physical Risk	Site-level exposure to 10 climate hazards under SSP1-2.6 through SSP5-8.5. Scored 0–100, maps to annual production loss % and Capital at Risk.	<i>35% of composite rating</i>
Transition Risk	Carbon price path, commodity demand shift, and technology disruption under NZE, Delayed Transition, and Current Policies scenarios. Scored 0–100.	<i>35% of composite rating</i>
Financial Impact	Revenue and EBITDA sensitivity, WACC uplift, Capital at Risk (CaR), and equity/debt impairment modelling. Calibrated to company financials.	<i>30% of composite rating</i>

2.2 Scenario Coverage

The engine natively supports the following internationally recognised climate scenarios:

SSP1-2.6 / NZE	Net Zero by 2050. ~1.5°C warming. High transition costs, low physical risk.	<i>IPCC AR6 · IEA NZE 2050</i>
SSP2-4.5 / DT	Delayed Transition. ~2.0°C warming. Moderate physical and transition risk.	<i>IPCC AR6 · NGFS Phase 4</i>
SSP3-7.0 / CP	Current Policies. ~2.7°C by 2050. High physical risk, low transition cost.	<i>IPCC AR6 · NGFS Phase 4</i>
SSP5-8.5 / HW	Hot House World. ~4.5°C. Severe physical damage, stranded assets.	<i>IPCC AR6 · NGFS Phase 4</i>

2.3 Time Horizons

Assessments are available at any target year between 2025 and 2100. Standard reporting horizons are 2030 (near-term), 2050 (medium-term), and 2100 (long-term), consistent with TCFD and IFRS S2 guidance.

3. Physical Risk Methodology

3.1 Hazard Coverage

The engine covers all 10 acute and chronic physical hazard categories identified by the IPCC AR6 Working Group II (Chapters 3–4) as financially material. All 10 are disclosed per TCFD, IFRS S2, and CSRD requirements.

Heat Stress	Wet-bulb globe temperature exceedance; labour productivity loss; operational shutdowns.	<i>IPCC AR6 WG2 Ch3.3</i>
Riverine Flood	Clausius-Clapeyron precipitation scaling; return period shifts; floodplain expansion.	<i>IPCC AR6 WG2 Ch4.2</i>
Coastal Flood	Storm surge compounded with sea level rise; saline intrusion; erosion risk.	<i>IPCC AR6 WG2 Ch3.4</i>
Sea Level Rise	Chronic inundation risk; asset stranding probability by elevation band.	<i>IPCC AR6 WG2 Ch3.4</i>
Saltwater Intrusion	Groundwater salinisation risk for coastal and low-elevation assets.	<i>IPCC AR6 WG2 Ch4.6</i>
Landslide	Terrain stability under increased precipitation intensity; slope and geology factors.	<i>IPCC AR6 WG2 Ch4.7</i>
Wildfire	Fire Weather Index (FWI) trend; NASA FIRMS historical ignition density overlay.	<i>IPCC AR6 WG2 Ch3.5</i>
Cyclone / TC	Tropical cyclone track intensification; IBTRACS historical path density.	<i>IPCC AR6 WG2 Ch11.7</i>
Drought	PDSI scaling; water availability stress; growing season disruption.	<i>IPCC AR6 WG2 Ch3.3</i>
Water Stress	WRI Aqueduct 4.0 baseline; demand-supply gap under warming and population growth.	<i>WRI Aqueduct 4.0</i>

3.2 Compound Risk Aggregation

Individual hazard severities are combined using a joint probability model that accounts for correlation and non-additivity between hazards. An asset exposed to both riverine flood and drought — for instance a brewery drawing on a flood-prone river in a semi-arid region — receives a compound score that reflects the interaction, not merely the sum of individual exposures.

The joint loss formula follows standard actuarial practice for correlated perils. The specific parameterisation is proprietary and available for review under NDA.

3.3 Site-Level Precision

When an asset's GPS coordinates are provided, the engine applies a multi-stage site physics layer on top of the regional climate baseline:

- Coastal distance correction: hazards including coastal flood, saltwater intrusion, and sea level rise are suppressed for assets more than 80km from the coastline or above 20m elevation, consistent with IPCC definitions of coastal zones.
- Elevation lapse rate: temperature-sensitive hazards (heat stress, wildfire) are adjusted for altitude using standard atmospheric lapse rates.
- Urban heat island effect: urban-classified assets receive an upward temperature adjustment reflecting documented UHI magnitudes.
- River proximity: riverine flood severity is amplified for assets within 5km of a major river channel.

- Slope factor: landslide applicability is assessed against terrain gradient; assets on flat terrain receive a not-applicable designation with explanatory note.

4. Spatial Downscaling — 25km Grid to Asset Level

4.1 The Downscaling Challenge

Global climate model projections (including NASA NEX-GDDP-CMIP6, the primary source for this engine) are produced at approximately 0.25° spatial resolution, equivalent to roughly 25km grid squares at mid-latitudes. This resolution is adequate for regional climate science but insufficient for asset-level financial risk assessment — a brewery, refinery, or port facility may sit in a very different microclimate from the centre of its 25km grid cell.

Naive approaches — simply assigning each asset the value of its nearest grid cell — introduce systematic bias, particularly for assets near coastlines, in river valleys, or at elevation boundaries.

4.2 The CRI Approach: GPS Bilinear Interpolation

The CRI Engine addresses this through a two-stage spatial downscaling process:

Stage 1 — Statistical baseline interpolation

Each asset's climate risk baseline is computed as the distance-weighted blend of its surrounding regional reference points. The weight assigned to each reference point decreases with the square of its distance from the asset — a standard technique in geostatistics known as inverse-distance weighting (IDW). Reference points more than a few hundred kilometres away receive negligible weight and do not influence the asset's score.

This approach is analogous to how weather services interpolate station observations to produce gridded weather maps — using nearby measurements weighted by proximity rather than assuming uniform values across large cells.

Stage 2 — Site physics adjustment

The interpolated regional baseline is then modified by the site-level physical factors described in Section 3.3 (elevation, coastal distance, urban heat island, slope). This second stage captures local geography that is invisible at any grid resolution.

4.3 Meteorological Data Foundation: ERA5 + CMIP6

All temperature and precipitation baselines are grounded in ERA5, the ECMWF's global reanalysis dataset covering 1940 to the present at approximately 31km resolution. ERA5 is the industry-standard observational baseline used by meteorologists, central banks, and climate scientists globally.

Future projections apply the "delta method" — the CMIP6 model's projected change signal is added to the ERA5 observational baseline rather than using raw model output. This avoids the mean-state bias that afflicts direct use of climate model temperatures, and is the approach recommended by the IPCC AR6 WG1 for applied climate risk assessments.

ERA5 Reanalysis	30-year historical temperature and precipitation baseline at exact asset GPS coordinates. Fetched on-demand for each asset.	ECMWF · Hersbach et al. 2020
NASA NEX-GDDP-CMIP6	Statistically downscaled daily climate projections 2015–2100. 9-model ensemble median applied via delta method.	NASA · Thrasher et al. 2022
IPCC AR6 WG1	Regional warming amplification factors and SSP scenario temperature ranges.	IPCC 2021 · SPM Table SPM.1

For each asset with GPS coordinates, the engine fetches the ERA5 30-year daily temperature and precipitation record at that exact location via the Open-Meteo ERA5 Archive API. This provides a genuine observational baseline — not a lookup table — updated on demand.

5. Primary Data Sources

All primary data sources are from peer-reviewed, publicly available, internationally recognised datasets. No proprietary third-party data purchases are required to run the engine.

WRI Aqueduct 4.0	Water stress, riverine flood, and drought baseline values for 80+ country/region codes.	<i>Kuzma et al. 2023</i>
NASA NEX-GDDP-CMIP6	Temperature and precipitation projections to 2100 under SSP1-2.6 through SSP5-8.5.	<i>Thrasher et al. 2022</i>
ECMWF ERA5	Historical reanalysis: temperature, precipitation, wind, humidity 1940–present.	<i>Hersbach et al. 2020</i>
IPCC AR6 WG1 & WG2	Regional warming patterns, hazard impact functions, scenario definitions.	<i>IPCC 2021, 2022</i>
SRTM / Copernicus DEM	Global 30m elevation data used for coastal distance and slope calculations.	<i>NASA / ESA</i>
NASA MODIS/HLS	Land use / land cover classification proxy for urban heat island and wildfire.	<i>NASA LP DAAC</i>
IBTRACS v04	Global tropical cyclone track database 1842–2024.	<i>Knapp et al. 2010, NOAA</i>
NASA FIRMS	Active fire radiative power density for wildfire baseline calibration.	<i>NASA EOSDIS 2024</i>
NGFS Phase 4	Carbon price pathways and scenario macroeconomic assumptions.	<i>NGFS 2023</i>

6. Regulatory Framework Alignment

The CRI Engine is designed to produce outputs that are directly usable for regulatory disclosure without significant reformatting. The table below maps each major framework requirement to the engine's output fields.

TCFD	Physical risk identification and disclosure; scenario analysis; financial impact quantification.	<i>All 10 IPCC-aligned hazards; SSP1-2.6 to SSP5-8.5; CaR and EV impact outputs</i>
IFRS S2	Climate-related risks and opportunities; cross-industry metric categories.	<i>Hazard severity scores; transition scenario analysis; Scope 1/2/3 emissions integration</i>
CSRD / ESRS E1	Double materiality assessment; physical and transition risk disclosure; biodiversity.	<i>Asset-level materiality flags; sector-specific impact functions; DNSH alignment</i>
ECB Guide	Physical risk heat mapping; transition risk sensitivity analysis; financed emissions.	<i>Portfolio-level CaR; loan book climate VaR; sector exposure heatmaps</i>
EBA Pillar 3	Template EU-5 quantitative climate risk exposure; concentration analysis.	<i>CRE and physical exposure tables; geographic concentration by hazard type</i>
ASRS E1	Australian Sustainability Reporting Standards physical risk disclosure.	<i>Full hazard coverage with AU-specific regional calibration</i>

7. Outputs and Deliverables

7.1 Standard Outputs per Asset

- Physical risk score (0–100) with per-hazard breakdown and severity ratings
- Annual production loss probability (joint probability across all applicable hazards)
- Capital at Risk (CaR) in GBP/USD/EUR, calibrated to asset replacement value or revenue exposure
- Enterprise Value impact under each scenario and time horizon
- Meteorological baseline: ERA5 historical mean, observed trend, and CMIP6 projection at target year
- Full audit trail: every score traces to a source citation, impact function reference, and methodology note

7.2 Portfolio-Level Outputs

- Climate Rating (0–100 composite) for each company or portfolio
- Sector-level risk concentration heatmap
- Transition cost trajectory (carbon cost as % EBITDA, 2026–2100)
- Supply chain vulnerability mapping across Tier 1 and Tier 2 suppliers
- Loan book climate VaR for financial institutions

7.3 Report Formats

- Interactive web dashboard (browser-based, no installation required)
- PDF disclosure report (TCFD / IFRS S2 / CSRD format)
- Excel data package with per-asset scores, sources, and audit trail
- Regulatory templates: ECB, EBA Pillar 3, ASRS pre-formatted

7A. Sample Output — Illustrative Case Study

What a CRI Engine Assessment Looks Like in Practice

The following illustrates a typical CRI Engine assessment output for a major European food & beverage company with production assets across multiple geographies. All figures shown are representative examples of engine output format and scoring methodology; they are not statements of the actual current climate risk of any named company.

Asset Profile: Central European Brewery · 51.69°N, 5.30°E · Netherlands · Food & Beverage

Physical Risk 62 MEDIUM-HIGH	Transition Risk 44 MODERATE	CaR (2050, SSP2) £ 8.3M Capital at Risk / yr	Prod. Loss Risk 4.1% Annual exposure by 2050
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Physical Hazard Breakdown (SSP2-4.5, Target Year 2050)

Riverine Flood	74	HIGH	Meuse/Rhine tributaries; 1-in-20yr event probability +38% by 2050
Water Stress	68	HIGH	WRI Aqueduct: NL-ZH baseline medium-high; demand gap widens under +2°C
Drought	61	MEDIUM	Summer PDSI projected -1.4 standard deviations by 2050; brewing water availability
Heat Stress	55	MEDIUM	WBGT exceedance days +14/yr by 2050; indoor cooling load uplift +18%
Extreme Precipitation	48	MEDIUM	Clausius-Clapeyron +7%/°C; 1-in-10yr intensity events become 1-in-7yr
Coastal Flood	12	LOW	Asset 34km inland; coastal flood not applicable at this location
Sea Level Rise	8	LOW	Asset elevation 6m ASL; inundation risk negligible to 2100
Wildfire	6	LOW	FWI baseline low for NL; temperate maritime climate; negligible
Cyclone / TC	4	LOW	Extra-tropical only; IBTrACS density near-zero for this region
Landslide	3	LOW	Flat terrain (<1° slope); geomorphologically stable; not applicable

Meteorological Baseline — Live ERA5 Data at Asset GPS

The following baseline values were fetched on-demand from the ECMWF ERA5 reanalysis archive at the asset's exact GPS coordinates. This is the observational foundation on which CMIP6 delta projections are applied.

Mean Temperature	10.6°C	+0.038°C/yr (ERA5)	+12.9°C (+2.3°C) ↗
Annual Precipitation	812 mm/yr	+1.8 mm/decade	891 mm/yr (+9.7%) ↗
Hot Days (>30°C)	7.4 days/yr	+0.4 days/yr	18.2 days/yr (+146%) ↗

Dry Days (<1mm)	214 days/yr	Stable	228 days/yr (+7%) ↗
Data Source	ERA5 · 10,957 days	open-meteo.com API	NASA NEX-GDDP-CMIP6

Scenario Comparison — Physical Risk Score Trajectory

SSP1-2.6 (Net Zero)	44	50	54	56	55
SSP2-4.5 (Delayed Trans.)	46	55	62	71	76
SSP3-7.0 (Current Policy)	48	58	67	79	88
SSP5-8.5 (Hot House)	51	63	74	87	96

How to Read This Output

The Physical Risk score of 62/100 under SSP2-4.5 by 2050 indicates a medium-high climate exposure. For a food & beverage producer, the primary material risks are water availability (direct production dependency on water volumes and quality) and riverine flood (asset damage and operational disruption). The £8.3M Capital at Risk figure represents the expected annualised climate-attributable loss in a medium-stress scenario — this maps directly to a TCFD financial disclosure line item.

Under SSP5-8.5 (no climate action), the score reaches 74 by 2050 and 96 by 2100 — approaching the "high climate risk" threshold that triggers regulatory heightened scrutiny under ECB and EBA guidance. This trajectory analysis is precisely what lenders, investors, and regulators need to see.

The scenario comparison table is the centrepiece of TCFD Section 4 (financial impact) and IFRS S2 cross-industry metric Category C. The engine produces this table automatically for any asset.

8. Limitations and Professional Caveats

Intellectual honesty about model limitations is a core principle of the CRI Engine. The following are clearly disclosed in all outputs:

- The engine is a parameterised quantitative framework — it is not a live geospatial API query or a bespoke physical hazard survey. It is calibrated to peer-reviewed science and suitable for regulatory disclosure support, but site-specific engineering surveys may be warranted for high-value assets.
- Precipitation projections carry substantially higher uncertainty than temperature projections ($\pm 8\%$ vs $\pm 0.3^\circ\text{C}$). Flood and drought outputs should be treated as indicative trend signals, not precise engineering estimates.
- The engine covers climate-related physical and transition risks. Geophysical hazards (earthquakes, volcanoes, tsunamis) are intentionally excluded from climate scores — they are not climate-sensitive and must not appear in TCFD/IFRS S2 climate disclosures.
- All outputs should be reviewed by a qualified risk professional before submission to regulators, lenders, or rating agencies.

Review by a qualified climate risk professional is recommended before regulatory submission. The CRI Engine is designed as a decision-support tool, not as a substitute for professional judgment.

9. About ClimRisk

ClimRisk is an independent climate risk intelligence platform founded to make institutional-grade physical and transition risk analysis accessible to companies and financial institutions of all sizes. The CRI Engine is built on open, peer-reviewed science — the same datasets and frameworks used by central banks, major rating agencies, and the IPCC itself — packaged for practical regulatory and investment use.

For partnership discussions, technical enquiries, or to arrange a live demonstration:

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climrisk.io/CRI_Demo.html

This methodology document is shared at Tier 2 confidence level for partnership evaluation purposes. Proprietary calibration details, exact model parameters, and technical implementation are available for review under a mutual NDA.